

Engagement and Equity as Evidence for AI-Informed Instructional Design: A Learning Analytics Study of Two Large-Scale Datasets

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Abstract

The rapid promotion of artificial intelligence as a vehicle for improving teaching and learning has outpaced the evidentiary base from which instructional designers might responsibly decide where such technology should intervene, leaving practitioners to adopt AI-enabled tools on the strength of technical promise rather than demonstrated leverage over student outcomes. The present study addresses this deficit by treating two of the largest publicly available learning analytics datasets not merely as sources of description but as evidence from which defensible design priorities can be derived, drawing jointly on the Open University Learning Analytics Dataset (OULAD; 28,785 students, 22 module presentations, 206 assessments) and a randomly sampled 4,000-user subset of EdNet-KT1, the interaction log of a deployed AI tutoring system (484,347 interactions). Across both datasets, behavioral engagement with course materials emerged as the strongest correlate of academic success observed in this study, exceeding in magnitude every structural feature of course design examined: virtual learning environment engagement was associated with passing at a point-biserial correlation of 0.48 ($p < .001$), while assessment load and course length showed weaker, though still significant, associations with outcomes ($r = -.44$ and $r = .48$, respectively). This behavioral signal, however, was distributed with marked inequality, both across the population of a live AI tutoring deployment, where engagement volume yielded a Gini coefficient of 0.86 and half of sampled users disengaged after a single session, and across socio-economically stratified subgroups of Open University students, among whom pass rates diverged by 22.4 percentage points between the most and least deprived deciles and by a further margin according to disability status (both $p < .001$). Read together, these findings indicate that an artificial intelligence agenda for instructional design achieves the greatest evidentiary support not by automating assessment or content delivery in isolation, but by treating early engagement, retention, and equity-disaggregated evaluation as first-order design targets that any adaptive or generative capability must be built to serve. Because this Collection is oriented explicitly toward United Nations Sustainable Development Goal 4, the equity evidence reported here bears directly on its aims, offering a disaggregated empirical account of where large-scale learning technologies presently fall short of inclusive, effective provision and where AI-informed design might close that gap rather than widen it.

1. Introduction

The proposition that artificial intelligence can improve teaching and learning has, over the past decade, moved from speculative claim to institutional expectation, yet the discourse surrounding this shift has been driven substantially by what AI can technically accomplish rather than by prior evidence of which instructional design levers most reliably move student outcomes. This asymmetry matters because instructional designers, teachers, and educational technologists are increasingly asked to embed AI into course design without a clear empirical account of where existing course structures already succeed, where they fail, and for whom they fail most severely. A recent systematic review of AI's contribution to higher education's quality-education and equity aims found that the evidence connecting specific AI applications to improved, equitably distributed outcomes remains considerably thinner than the volume of published enthusiasm for the technology would suggest [1], a conclusion consistent with a broader

review of the state of the art in AI in education, which found that much of the field's most consequential work still concerns profiling, prediction, and adaptive systems rather than the course-design decisions that determine what those systems have to work with [2]. The gap is not merely academic. A systematic review of learning-analytics adoption across higher education institutions found that although most institutions recognize considerable potential in learning analytics for informing course design and student support, adoption remained uneven and was constrained less by interest than by the organizational, infrastructural, and professional-development conditions surrounding these systems [3]. It is precisely this gap, between technological aspiration and design-actionable evidence, that the present study is constructed to close, and it corresponds closely to the aim this Collection sets for itself: bringing advanced technologies and innovative teaching together in a manner instructional designers can put into practice.

Closing that gap requires an evidence-first posture rather than a tool-first one. Rather than proposing a new AI application and testing its efficacy in isolation, this study poses a prior and, we argue, more consequential question: across large populations of real students, which structural and behavioral signals are already associated with success or failure, and which of those signals are ones that an AI-informed instructional design process could plausibly act upon? Answering this question at the scale required to support generalizable claims is difficult to achieve through small classroom studies alone, which is why we turn to two openly licensed, large-scale learning analytics datasets whose combination spans both a conventional virtual learning environment operating at institutional scale and a purpose-built AI tutoring system operating at commercial scale.

The Open University Learning Analytics Dataset (OULAD) records course structure, assessment design, demographic composition, and virtual-learning-environment interaction for 28,785 students across seven distance-learning modules at the UK Open University [4]. EdNet, for its part, is the interaction log of Santa, an AI-driven tutoring service for English-language test preparation used by more than 780,000 students, and constitutes, to its creators' knowledge, the largest publicly available record of student-AI-tutor interaction [5]. OULAD has already supported research on learning design and predictive analytics at the institution from which it originates, including a recent multi-institutional finding that the specific design of a course, more than instructor ratings, predicts both students' evaluation of course quality and their behavioral engagement with it [6], and evidence that a university-wide predictive analytics system built on data of this kind changed teaching practice at scale several years into its deployment [7]. The present study extends this line of inquiry by explicitly connecting course-design and engagement evidence drawn from a conventional virtual learning environment to engagement evidence drawn from a live, deployed AI tutoring system, so that the resulting argument speaks directly to instructional designers who must decide how, and where, to integrate AI into course design. It also bears directly on the broader project of connecting artificial intelligence to Sustainable Development Goal 4, quality education, by grounding that project in disaggregated, empirical equity outcomes rather than in aspiration alone [8].

This evidence-first posture is, if anything, more urgent now than when OULAD and EdNet were first collected, because the technology instructional designers are being asked to integrate has itself changed. Generative AI and large language models have, since 2023, introduced a further layer of institutional expectation on top of the adaptive and predictive systems this paper analyzes, and recent scholarship on their arrival in higher education emphasizes both their capacity to generate diverse, context-relevant instructional content and the considerable uncertainty that remains about how that capacity should be governed by instructional design principles rather than by the technology's own momentum [9]. Emerging work on generative-AI-enhanced instructional design has begun to propose co-creative models in which the designer, rather than the algorithm, retains authority over pedagogical decisions while generative tools handle drafting and iteration, a division of labor that presupposes exactly the kind of evidence about design leverage that this paper sets out to provide [10]. Framed this way, the present analysis is not an argument against generative AI or adaptive tutoring; it is an argument that whichever AI capability an instructional designer chooses to integrate, generative or otherwise, its deployment should be organized around the same evidentiary priorities that a purely data-driven analysis of course design and engagement independently recommends.

Three questions organize the analysis that follows. The first concerns course design: do observable choices such as the number and type of assessments, course length, and deadline structure correlate with student outcomes in ways that could inform AI-assisted, design-time decision support? The second concerns engagement and equity: how much of the variance in outcomes is associated with behavioral engagement, and does that engagement vary systematically by demographic characteristics in ways directly relevant to Sustainable Development Goal 4? The third concerns adaptivity itself: in a system where AI already personalizes practice, namely Santa within EdNet, what do observed engagement patterns imply about the design problem instructional designers actually face when embedding adaptive AI into a course, is the central challenge one of content adaptivity, or one of sustaining engagement long enough for adaptivity to have any opportunity to matter? Section 2 develops the theoretical framework from which these questions follow; Section 3 details data and methods; Section 4 reports results organized around the three questions; and Sections 5 through 7 draw out implications for AI-informed instructional design practice, for equity, and for future research.

2. Theoretical Framework and Related Work

2.1 Instructional design theory and the role of course structure

Instructional design theory has long held that the structural features of a course, the sequencing and pacing of content, practice, and assessment, are not incidental to learning but among its primary determinants. Merrill's synthesis of first principles of instruction, revisited and reaffirmed in a recent handbook treatment of the theory, argues that learning is promoted when learners engage with authentic problems, activate prior knowledge, are shown new knowledge, apply it with feedback, and integrate it

into their own practice, a sequence that is a property of course design rather than of any particular delivery technology [11]. A recent systematic review and meta-analysis of scaffolding through prompts in digital learning environments reaches a compatible conclusion from a different direction, finding that structured, prompt-based guidance reliably improves learning achievement relative to unguided digital practice, and that the benefit is largest precisely where learner guidance is weakest without it [12]. Taken together, these positions predict that measurable features of course structure, assessment frequency and type, pacing, and course length among them, should exhibit systematic relationships with outcomes independent of any AI layer subsequently added to a course. Section 4.2 tests this prediction directly against OULAD's module-level structural data.

2.2 Learning analytics, engagement, and learning design

A parallel literature in learning analytics has established that behavioral engagement with course materials, and not prior achievement alone, predicts persistence and performance in online and blended courses. A cross-institutional study drawing on the same UK distance-learning context examined here found that the specific design of a course predicts students' evaluated quality and engagement more strongly than instructor ratings do, underscoring that engagement is substantially a design outcome rather than solely a student or teacher attribute [6]. A recent systematic literature review of how learning analytics has been used to improve learning design concluded that the field has moved from simply describing engagement after the fact toward using analytic evidence to revise course design before problems recur, though the review also cautioned that this improvement cycle remains inconsistently operationalized across institutions [13]. Complementing this design-level evidence, work applying interpretable machine learning to academic-achievement prediction has shown that models built for a specific course or context substantially outperform generic predictors transferred from elsewhere, reinforcing the point that engagement and its predictive value are locally, not universally, determined [14]. Self-regulated learning offers a further behavioral lens on the same phenomenon: a systematic review of studies using learning analytics to measure self-regulation in higher education found that indicators of time management, regularity, and sustained practice, the same class of signal examined in this paper's engagement analysis, are among the most consistent behavioral correlates of self-regulated learning identified in the literature [15]. These findings, read together, imply that engagement is not merely a symptom of student ability but is itself substantially shaped by design choices, a premise that motivates the analysis of OULAD's virtual-learning-environment clickstream data undertaken in Section 4.3 and visualized in Fig. 3.

2.3 Intelligent tutoring systems and AI-driven personalization

A third and longer-standing literature has examined whether adaptive, computer-based tutoring can approximate the benefits of individualized human instruction. A recent systematic review of AI-driven intelligent tutoring systems found generally positive, though variable, effects on learning and performance across the studies examined, with the size of the benefit depending heavily on how directly the tutoring system's content aligned with what was subsequently assessed [16]. More recent work has

continued to advance the underlying modeling techniques on which such systems depend: deep-learning approaches to knowledge tracing, which estimate a learner's evolving mastery from sequences of item responses, have improved the accuracy with which systems such as the one underlying EdNet can, in principle, model what a student does and does not yet know [17]. What this literature has been comparatively less able to establish, because intelligent-tutoring-system efficacy studies are typically conducted with motivated, retained samples, is how personalization performs under the real-world conditions of voluntary use, attrition, and highly unequal engagement that a full-scale commercial deployment necessarily contains, conditions this study is able to observe directly in EdNet's deployment data and interprets in Section 4.6 and Section 5.4.

2.4 Equity as a design requirement

Equity has become a central and explicit concern in both learning analytics and AI-in-education scholarship, reflecting recognition that automated or data-driven systems can reproduce or amplify existing disparities where they are not evaluated explicitly for differential impact across student populations. A recent systematic review of the theoretical underpinnings of online-education attrition research argues that a coherent theory of retention specific to online and distance settings remains underdeveloped relative to the volume of predictive modeling built on top of it, so that attrition models risk optimizing prediction without a clear account of what, structurally, is being predicted [18]. Directly relevant to the datasets examined here, a recent fairness audit of machine learning models trained on OULAD found measurable bias across demographic subgroups in standard predictive models, and showed that bias-mitigation techniques can reduce these disparities while preserving acceptable predictive accuracy [19]. A broader systematic mapping of fairness research across educational machine learning corroborates this pattern, finding that gender and disability are among the sensitive attributes most commonly examined, and that fairness is frequently traded off against accuracy rather than pursued as a joint design objective [20]. The concern extends well beyond OULAD: a large-scale audit of predictive algorithms used across United States higher education found that such models systematically underestimated the likelihood of success for Black and Hispanic students while overestimating it for White and Asian students, with false-negative rates for Black and Hispanic students roughly double those observed for White and Asian students [21]. This body of scholarship motivates the explicit two-way equity analysis undertaken in Section 4.5 and visualized as a heatmap in Fig. 2, and grounds the argument developed in Section 5.3 that equity is properly understood as a design requirement for AI-informed instructional design, rather than as an afterthought appended once a system has already been deployed.

3. Methods

3.1 Data sources

OULAD contains data for seven course modules, spanning 22 module presentations in total (a module as offered in a specific term) at the Open University in the United Kingdom, comprising 28,785 students,

206 assessments, and 173,912 individual assessment-submission records [4]. Course-structure data include, for each module presentation, the number and type of assessments (tutor-marked assessment, computer-marked assessment, and final examination), their assigned weight, and submission deadlines, together with module length in days. Student-level data include demographic characteristics (gender, self-declared disability status, an Index of Multiple Deprivation band reflecting the socio-economic profile of the student's home area on a ten-decile scale, highest prior educational qualification, and age band), registration date relative to module start, and final module outcome (Distinction, Pass, Fail, or Withdrawn). A separate and considerably larger table (studentVle.csv, comprising approximately 10.6 million rows) records time-stamped counts of student clicks on each of twenty types of virtual-learning-environment material, among them course content pages, forum posts, quizzes, and the module homepage. All data are anonymized and released under a Creative Commons Attribution 4.0 licence.

EdNet-KT1 is the question-response interaction log of Santa, a multi-platform AI tutoring service for English-language test preparation, comprising 784,309 per-student files and more than 131 million interactions in total, released under a Creative Commons Attribution-NonCommercial 4.0 licence [5]. Each interaction record includes a timestamp, a question identifier, the student's selected answer, and elapsed response time in milliseconds. Given the scale of the full release, approximately 3 gigabytes compressed and 5.6 gigabytes uncompressed across 784,309 files, we drew a simple random sample of 4,000 of these files (0.51 percent), yielding 484,347 interactions, a sample large enough to estimate population-level distributional statistics with high precision while remaining computationally tractable. The item-level answer key required to score response accuracy, linking question identifiers to correct answers, skill tags, and item bundles, could not be fully retrieved within the scope of this study; the EdNet analysis is therefore restricted to interaction-level engagement and timing metrics, which do not depend on the answer key and which the wider learning-analytics literature treats as informative proxies for practice intensity, pacing, and behavioral automaticity [13].

3.2 Variable construction

From OULAD, we constructed a binary pass indicator (Distinction or Pass coded 1; Fail or Withdrawn coded 0) and a corresponding binary withdrawal indicator at the student level. At the module-presentation level, we computed the total number of assessments, the count of each assessment type, and the pass and withdrawal rates aggregated across all students within that presentation. Student-level virtual-learning-environment engagement was computed by summing recorded clicks across all interaction types for each student within their module presentation, using chunked aggregation across the full 10.6-million-row clickstream table; students with no recorded clicks were retained with a value of zero rather than excluded, so that the resulting distributions, reported as a boxplot in Fig. 3, reflect the full population rather than only its engaged subset. Assessment scores were merged with assessment metadata to compute, for every submission, the assessment type, the interval in days between submission and deadline (days late, with negative values indicating early submission), and score. For the equity analysis, we additionally cross-tabulated final outcome by both socio-economic deprivation decile and disability status jointly, rather than examining each demographic dimension in isolation, since the

compounding of disadvantage across dimensions is itself an empirical question this dataset can address directly (Section 4.5, Fig. 2).

From EdNet-KT1, for each sampled user we computed the total number of interactions, mean and median elapsed response time, and, after sorting interactions chronologically, the number of distinct practice sessions, defining a new session as beginning whenever the gap between consecutive interactions exceeded thirty minutes, a threshold consistent with conventional session-segmentation practice in web and clickstream analytics. From session counts we derived interactions per session and, from the first and last interaction timestamps, an engagement span in days. We additionally computed the Gini coefficient of interaction counts across sampled users as a summary measure of how unequally engagement volume was distributed, and constructed a Lorenz curve to visualize this inequality directly (Fig. 5).

3.3 Analytic approach

Pearson correlation was used for continuous module-level associations, for example assessment count against pass rate, and point-biserial correlation for associations between a continuous variable, virtual-learning-environment clicks, and a binary outcome, passing. Equity analyses used chi-square tests of independence between demographic categories, disability status and Index of Multiple Deprivation band, and the four-category final outcome; the joint relationship between deprivation and disability is additionally visualized as a two-way heatmap of pass rate in Fig. 2, since the two forms of disadvantage compound rather than operate independently. Assessment-type comparisons used descriptive statistics on score and on submission lateness. For EdNet, Pearson correlation related engagement-volume measures to timing measures, and Welch's t-test, which does not assume equal variances, compared mean elapsed response time between the bottom and top quartiles of user engagement volume. All analyses were conducted in Python 3.10 using pandas 2.3.3, NumPy 2.2.6, and SciPy 1.15.3; figures were produced with Matplotlib.

3.4 Scope and data availability limitations

This is a correlational secondary data analysis, and no causal claims are warranted by it. OULAD reflects one UK institution's distance-learning courses from 2013 to 2014, and its computer-marked-assessment category reflects automated grading rather than adaptive, model-driven assessment, a limitation when the dataset is used as a proxy for contemporary generative-AI-based instructional design. That said, a recent cross-national meta-analysis found blended and distance-learning formats broadly comparable to this one to produce consistent, positive effects on performance, attitude, and engagement across a range of countries and educational systems, which lends some support to treating OULAD's distance-learning context as informative beyond the United Kingdom specifically, even though it cannot substitute for direct replication [22]. The EdNet analysis is restricted to a random sample of the full KT1 release for computational tractability, and excludes response-accuracy and knowledge-tracing measures because the complete item answer key was not accessible within the scope of this study. Both limitations are addressed further, alongside others, in Section 6.

4. Results

4.1 Sample characteristics

The OULAD sample comprised 28,785 students across seven modules and 22 module presentations, with a median of 1,318 students per presentation (range 383 to 2,498). Across the full sample, students most commonly passed (37.9 percent) or withdrew (31.2 percent), with 21.6 percent failing and 9.3 percent achieving Distinction (Table 1, Fig. 1). The EdNet sample comprised 4,000 randomly selected users, 0.51 percent of the full 784,309-user KT1 release, contributing 484,347 individual question-response interactions.

Table 1
Final outcome distribution, OULAD (N = 28,785 students).

Outcome	n	Percentage of students
Pass	12,361	37.9%
Withdrawn	10,156	31.2%
Fail	7,052	21.6%
Distinction	3,024	9.3%

4.2 Course and assessment structure are associated with outcomes

At the module-presentation level ($n = 22$), the total number of assessments correlated negatively with pass rate ($r = -.44$) and weakly positively with withdrawal rate ($r = .16$), while module length in days correlated positively with pass rate ($r = .48$) and negatively with the total number of assessments ($r = -.34$). In this dataset, then, longer modules tended toward fewer, more evenly spread assessments and toward higher pass rates, a pattern directly consistent with the course-design-as-predictor finding reported elsewhere in this same institutional context [6]. The proportion of assessments that were computer-marked bore no statistically significant relationship to pass rate ($r = -.20$, $p = .37$); with only 22 module presentations available for this comparison, the finding is underpowered and should be read as inconclusive rather than as evidence of a genuine absence of effect.

Scored assessment type also differed systematically (Table 2, Fig. 4). Mean scores were highest on computer-marked assessments (mean 81.0, median 82, $n = 70,527$), followed by tutor-marked assessments (mean 72.6, median 76, $n = 98,253$) and final examinations (mean 65.6, median 67, $n = 4,959$). Computer-marked assessments also carried a far higher rate of late submission, 46.8 percent, than tutor-marked assessments (16.6 percent) or examinations (0.1 percent), and later submission was weakly though significantly associated with lower scores overall ($r = -.17$, $p < .001$). Together these patterns are most plausibly read as evidence that computer-marked components functioned as flexible,

resubmittable formative practice rather than as strictly time-boxed summative assessment, a distinction that becomes important when we return, in Section 5.2, to how such components might be extended through AI-generated practice items.

Table 2
Score and submission timing by assessment type, OULAD.

Assessment type	Mean score	Median score	SD	n submissions	Percentage submitted late
Computer-marked (CMA)	81.0	82	17.8	70,527	46.8%
Tutor-marked (TMA)	72.6	76	18.4	98,253	16.6%
Exam	65.6	67	20.6	4,959	0.1%

4.3 Engagement is the strongest behavioral correlate of success

Total virtual-learning-environment engagement, summed clicks across all course materials, showed the strongest association with passing of any variable examined in this study (point-biserial $r = .48$, $p < .001$), markedly exceeding any single course-structure variable reported in Section 4.2. The full distribution of engagement by outcome, rather than its median alone, is shown as a boxplot on a logarithmic scale in Fig. 3, a representation chosen deliberately because the underlying distributions are heavily right-skewed and overlap substantially between adjacent outcome categories, a texture that a bar chart of central tendency would otherwise conceal. The medians nonetheless rose monotonically with outcome, from 89 clicks among Withdrawn students to 317 among students who failed, 1,343 among students who passed, and 1,896 among students who achieved Distinction (Table 3), a greater than twenty-one-fold difference in median engagement between the lowest- and highest-performing groups, even as Fig. 3 makes clear that the interquartile ranges for Pass and Distinction overlap considerably, indicating that engagement, while strongly associated with success, is not by itself a sufficient condition for it. Course content pages received the largest share of total clicks (11.2 million), followed by the discussion forum (8.0 million), formative quizzes (7.0 million), and the module homepage (6.9 million), indicating that engagement with core content and peer discussion, and not merely assessment attempts, distinguished more successful students.

Table 3
Total VLE clicks by final outcome, OULAD.

Outcome	Median clicks	Mean clicks
Withdrawn	89	314.0
Fail	317	651.9
Pass	1,343	1,921.8
Distinction	1,896	2,666.8

4.4 Registration timing and early commitment

Registration date, recorded relative to module start, with negative values indicating registration before the start date, showed a counter-intuitive pattern: students who ultimately withdrew had registered earliest, on average (median minus 67 days before start), compared with students who passed (median minus 53 days), failed (median minus 50 days), or achieved Distinction (median minus 52 days). Rather than signalling stronger commitment, early registration in this dataset was associated with a higher eventual withdrawal rate, a pattern we treat as descriptive rather than causal. One plausible interpretation is that students who register furthest in advance include a subgroup less certain of their capacity to commit and more prone to early life-event withdrawal, but the dataset cannot adjudicate between this and competing explanations, for instance differences in funding route or module type among early registrants, and statistical power sufficient to test such explanations directly was not available here.

4.5 Substantial, statistically significant equity gaps in outcomes

Pass rates differed by disability status, 48.2 percent for students without a declared disability against 38.2 percent for students with a declared disability, a 10.0 percentage-point gap (chi-square with 3 degrees of freedom equals 138.45, $p < .001$), and, more sharply still, by neighborhood-level socio-economic deprivation, with students from the most deprived decile passing at 35.2 percent against 57.5 percent among students from the least deprived decile, a 22.4 percentage-point gap (chi-square with 27 degrees of freedom equals 666.94, $p < .001$; Table 4). Rather than treating these two forms of disadvantage as independent, Fig. 2 presents them jointly as a heatmap of pass rate across deprivation decile and disability status, and the resulting pattern is compounding rather than merely additive: the most deprived students with a declared disability passed at only 30.1 percent, some 28 percentage points below the least deprived students without a declared disability (58.1 percent), a gap considerably wider than either dimension of disadvantage produces on its own. Gaps by gender were comparatively small (48.4 percent for women against 46.2 percent for men). Prior educational attainment showed the widest gradient of any demographic variable examined, 29.7 percent among students with no formal qualifications on entry against 65.5 percent among students entering with a postgraduate qualification, a

35.8 percentage-point difference. These patterns are consistent with, and extend to a large UK distance-learning population, audits finding that predictive models trained on OULAD carry measurable demographic bias unless explicitly corrected [19], and they echo, in a UK context, the racialized disparities in predictive accuracy documented across United States institutions [21].

Table 4
Pass rate by socio-economic deprivation
decile (Index of Multiple Deprivation
band), OULAD.

IMD band (decile)	Pass rate
0–10% (most deprived)	35.2%
10–20%	38.6%
20–30%	40.8%
30–40%	46.9%
40–50%	46.6%
50–60%	48.8%
60–70%	51.9%
70–80%	51.5%
80–90%	54.1%
90–100% (least deprived)	57.5%

4.6 Engagement in a deployed AI tutoring system is highly unequal and short-lived

In the 4,000-user EdNet-KT1 sample (484,347 interactions), the distribution of engagement volume was extremely skewed (skewness = 11.7): the mean number of interactions per user (121.1) exceeded the median (11.0) more than tenfold, and the top decile of users, those at or above the 90th percentile with 184 interactions, accounted for a disproportionate share of total practice volume, yielding a Gini coefficient of 0.86 for interaction counts across users (Fig. 5, Fig. 6), a level of inequality comparable in magnitude to income-inequality figures reported for some of the world's most unequal economies, here applied instead to engagement with an AI tutoring system rather than to income. Half of sampled users generated a session count of one (median sessions per user = 1.0; mean = 6.9), and median engagement span was zero days, meaning that at least half of sampled users interacted with the system on a single day only, though a substantial tail persisted much longer (90th-percentile span = 65.7 days; Table 5).

Mean elapsed response time per question was approximately 25.3 seconds (median of user medians = 21.0 seconds). Users in the top quartile of engagement volume answered marginally but significantly

faster, on average, than users in the bottom quartile (mean 25,422 milliseconds against 26,818 milliseconds; Welch's $t = 2.61$, $p = .009$), a pattern consistent with a modest fluency or automaticity effect arising from repeated practice. Total interaction volume correlated moderately with engagement span ($r = .43$, $p < .001$), confirming that sustained use, rather than session frequency alone, drives cumulative practice; the number of sessions correlated only weakly with mean elapsed time ($r = .07$, $p < .001$), a statistically significant but practically small effect given the very large sample size underlying the test.

Table 5
Engagement descriptive statistics, EdNet-KT1 sample
($n = 4,000$ users).

Metric	Value
Users sampled	4,000
Total interactions in sample	484,347
Mean interactions per user	121.1
Median interactions per user	11.0
Gini coefficient (interaction volume)	0.86
Median sessions per user	1.0
Median engagement span (days)	0.0
90th-percentile engagement span (days)	65.7
Mean elapsed time per question (ms)	25,287

5. Discussion

5.1 Reconciling course design evidence with instructional design theory

The module-level associations reported in Section 4.2, more assessments associated with lower pass rates, longer modules associated with higher pass rates, sit comfortably within instructional design theory's longstanding emphasis on structure and pacing rather than on sheer assessment volume [11, 12], and echo prior OULAD-adjacent findings that the specific pedagogical design of a module, and not student characteristics alone, measurably shapes behavior and outcomes [6, 14]. Read alongside that literature, our results should not be interpreted as fewer assessments causing higher pass rates in isolation; rather, assessment count in this dataset appears to co-vary with module length and, plausibly, with pacing and workload distribution, which prior learning-design research identifies as the more proximate design lever [6]. The distinction carries real weight for how AI-informed instructional design ought to use evidence of this kind: a recommendation engine that simply advised reducing assessment count, without attending to pacing and workload distribution, would very likely misidentify the

mechanism actually at work, and would risk optimizing a proxy rather than the underlying design lever it was meant to represent.

5.2 Implications for AI-informed instructional design practice

Taken together, the results reported in Section 4 point toward concrete, prioritized design targets rather than toward a generic case for adding AI wherever it can technically be inserted, and five such priorities follow directly from the evidence, ordered here by the strength of the support each receives. The first and best-supported priority is that AI applications intended to sustain engagement warrant precedence over applications aimed principally at automating grading, wherever institutional resources are limited: the engagement-outcome association observed in Section 4.3 ($r = .48$) exceeded by more than double the largest course-structure association reported in Section 4.2 ($r = -.44$), and far exceeded the non-significant association between computer-marked-assessment share and pass rate ($r = -.20$, $p = .37$), so that, at least within this evidence base, sustaining behavioral engagement constitutes a substantially larger lever on outcomes than does the specific mix of automated versus tutor-marked assessment. The second priority follows from the same body of evidence read prospectively rather than retrospectively: because assessment structure and course length were measurably associated with outcomes, AI is better positioned as design-time decision support than as a run-time personalization layer alone, since models trained on datasets of this kind could plausibly simulate how a proposed change to assessment count, mix, or pacing would shift predicted pass and withdrawal rates before a course is deployed, moving AI earlier into the design process, alongside the instructional designer, rather than confining it to a role exercised only after launch.

A third priority concerns the formative function that automated assessment already appears to serve. Computer-marked assessments in this dataset carried both the highest average scores and, at 46.8 percent against 16.6 percent for tutor-marked work, by far the highest rate of late submission, a combination consistent with flexible, formative use rather than strictly time-boxed summative assessment. Instructional designers might reasonably extend this pattern with AI-generated low-stakes practice items, provided that deadline design does not inadvertently penalize the very flexibility that appears to support learning. A fourth priority follows from Section 4.6 rather than from OULAD: because at least half of users in a live AI tutoring deployment disengage after a single session, well before an adaptive algorithm has accumulated enough data to personalize meaningfully, onboarding and early retention constitute a design problem properly distinct from adaptivity itself, and design effort directed at the first one or two sessions of an AI-supported learning experience is likely to carry more leverage over whether personalization can help a given student than design effort directed at refining the underlying adaptivity algorithm. The fifth and final priority is procedural rather than substantive: equity-disaggregated evaluation should be built into any AI-informed design initiative from its outset, per the argument developed in Section 5.3, rather than added retrospectively once disparities have already been observed, a recommendation that follows directly from recent fairness-auditing findings concerning this same population of learners [19, 20].

5.3 Equity as a design requirement, not an afterthought

The 22.4 percentage-point pass-rate gap by neighborhood deprivation, and the compounding gap of some 28 percentage points once disability status is considered jointly with deprivation in Fig. 2, are large enough to be practically significant and not merely statistically so, and they mirror precisely the equity concerns that motivate research connecting artificial intelligence to Sustainable Development Goal 4, namely that education be inclusive and effective for all learners rather than only for those already well served [8]. A systematic review of the theoretical foundations of online-education retention argues that attrition is properly understood as an outcome of the interaction between student background and institutional structures of academic and social integration, rather than as a matter of individual deficit [18], and our finding that Withdrawn students registered earliest, on average, rather than latest (Section 4.4), is consistent with attrition being shaped by structural rather than purely dispositional factors, though we stop short of asserting a causal mechanism.

AI-driven personalization is frequently promoted as a route toward equity, yet personalization that is not explicitly evaluated for its effect on gaps of this kind risks reproducing rather than closing them. A recent audit of machine learning models trained on OULAD found systematic bias by disability status, with high overall accuracy failing to guarantee equitable predictions for disabled students, and showed that post-processing bias mitigation can reduce these disparities, generally at a modest cost to overall accuracy [19]. This is not a peculiarity of the Open University's data: a large-scale audit of predictive algorithms used across United States higher education institutions found comparable, and in some respects more severe, disparities, with models incorrectly predicting failure for Black and Hispanic students at roughly double the rate observed for White and Asian students, while simultaneously overestimating success among the latter groups [21]. A broader systematic mapping of fairness research across educational machine learning found the accuracy-fairness trade-off documented in both studies to be common across the field, and found gender and disability to be among the sensitive attributes most frequently examined [20]. On the strength of this convergent evidence from two different countries and two different demographic framings, we suggest that any AI-informed instructional design initiative building on the findings reported here should report equity-disaggregated outcomes, by disability status and by deprivation decile at minimum and, where the two can be examined jointly as in Fig. 2, in combination, as a standard component of evaluation, treating the accuracy-fairness trade-off documented in this literature as a design parameter to be set deliberately rather than as an unintended side effect discovered after deployment.

5.4 What a deployed AI tutoring system reveals about the adaptivity-retention tradeoff

The EdNet results reframe, rather than simply extend, the design problem that AI in instructional design must solve. Santa's underlying tutoring approach can, in principle, adapt content to a learner's estimated knowledge state through knowledge-tracing techniques whose accuracy continues to improve [17], consistent with the general promise documented in the intelligent-tutoring-systems literature [16], yet the

engagement data examined in Section 4.6 show that adaptivity has very little opportunity to operate for most users. A median of one session and a median engagement span of zero days together mean that at least half of sampled users disengaged before a second session occurred, well before enough interactions had accumulated for fine-grained adaptation to matter. The extreme inequality in engagement volume, a Gini coefficient of 0.86 as shown in Fig. 5, indicates further that a comparatively small set of highly engaged users generates most of the data from which any adaptive system can learn, and most of the practice from which any student stands to benefit.

For instructional designers, the implication is that the first-order design problem in embedding an AI tutor is not content adaptivity but onboarding and early retention, so that design effort spent on the first one or two sessions is likely to carry more leverage over whether AI personalization can help a given student than design effort spent refining the adaptivity algorithm itself. This conclusion is consistent with, and extends into an AI-tutoring context, the OULAD finding that engagement itself, more than any single structural or automation feature, is the dominant behavioral signal (Section 4.3). The modest fluency effect we observed, faster mean response times among higher-engagement users, is instructive in this light: it is consistent with returning practice supporting automaticity, and offers one small piece of evidence that the interactions which do occur are not merely repeated failure or random guessing, but it is far too small an effect, observed on far too select a sample of surviving engaged users, to be read as evidence that the tutoring system is broadly effective for the majority who disengage early.

5.5 Toward a practical design framework

Synthesizing Sections 5.1 through 5.4, we propose that AI-informed instructional design proceed through three sequential design questions, rather than beginning from the question of where an AI feature can be added. The first concerns retention: will this design carry a representative cross-section of intended learners past their first one or two sessions, given that engagement, more than any single structural feature, was the strongest correlate of success observed in this study? The second concerns structure: does the pacing, assessment mix, and workload distribution reflect established instructional design principles [11, 12], independent of whatever AI layer is subsequently added? The third concerns equity: has the design been evaluated separately, and where relevant jointly, for learners with disabilities and for learners from more deprived backgrounds, rather than only on average across the full population? Only once these three questions have been addressed does the specific choice of AI technique, adaptive sequencing, automated grading, generative content, or predictive early warning, become the primary design decision, a sequencing that inverts the order in which AI adoption is often currently pursued.

5.6 Where generative AI fits within this evidence base

Because this Collection addresses emerging technologies as well as established ones, it is worth being explicit about how the present findings bear on generative AI specifically, since neither OULAD nor EdNet contains any generative-AI interaction and the temptation to over-extend behavioral findings from earlier technologies to a qualitatively different one should be resisted. Recent scholarship on large language models in higher education frames their most defensible role as a supplementary instructional resource

whose educational value depends on tight alignment with course objectives and instructional design, rather than as a freestanding substitute for course design itself [9], a framing that sits comfortably alongside this paper's central finding that structural and behavioral design levers, not the presence or absence of any particular automation technology, are what most reliably distinguish successful courses from unsuccessful ones. Proposals for AI-enhanced instructional design that explicitly preserve the designer's authority over pedagogical decisions, using generative tools for drafting and iteration while keeping design judgment human, are consistent with the three-question framework proposed in Section 5.5: a generative-AI-assisted course still needs to pass the retention, structure, and equity tests before its use of generative content becomes the primary design decision, and there is no evidence in the datasets examined here that generative capability changes what those three tests should look for [10]. If anything, the extreme early disengagement documented in Section 4.6 suggests that generative AI's much-discussed capacity for personalized content generation will matter for student outcomes only to the extent that it is paired with an equally serious design effort aimed at the first one or two sessions of use, since a highly capable generative tutor that most students abandon before it can adapt to them offers no more practical benefit than the comparatively simpler adaptive system examined in this paper.

6. Limitations

Several limitations qualify these findings. First, both datasets are observational, and the associations reported here, between engagement and passing, for instance, cannot be interpreted causally; more capable or more motivated students may both engage more and perform better for reasons unrelated to course design itself. Second, OULAD reflects a single UK institution's 2013-to-2014 distance-learning modules; its computer-marked-assessment category is automated but not adaptive or generative-AI-based, which limits how directly its findings transfer to contemporary generative-AI instructional design, and its population of adult distance learners may not generalize to campus-based or K-12 contexts. Third, module-level structural correlations reported in Section 4.2 rest on only 22 module presentations, yielding limited statistical power; the non-significant computer-marked-assessment-share finding in particular should be treated as inconclusive rather than as evidence of no effect, and the registration-timing finding in Section 4.4 is descriptive and could reflect unmeasured confounds, for example funding route or module type, rather than a general phenomenon.

Fourth, the EdNet analysis uses a random 0.51 percent sample, 4,000 of 784,309 users, of KT1 for computational tractability; while random sampling supports unbiased point estimates of the distributions reported, rare behavioral patterns may be underrepresented, and the sample, drawn from a single commercial AI tutoring product for English-language test preparation, may not generalize to other AI tutoring contexts or subject domains. Fifth, and most consequentially, the complete EdNet item-level answer key was not accessible within the scope of this study, so no response-accuracy or knowledge-tracing measures are reported; the engagement and timing findings should accordingly be read as evidence about behavioral persistence and pacing rather than about learning gains or mastery. Sixth, the equity analysis in OULAD relies on the categories available in the dataset, a binary disability flag and a neighborhood-level deprivation decile, both of which are coarse proxies for the underlying constructs of

disability and socio-economic disadvantage and cannot capture within-category heterogeneity. Seventh, and most relevant to this Collection's emphasis on emerging technologies, neither dataset contains any generative-AI interaction, so Section 5.6's discussion of generative AI is necessarily an extrapolation from behavioral principles rather than a direct empirical finding, and should be weighted accordingly relative to the dataset-grounded claims that precede it.

7. Future Research

Three extensions follow directly from the limitations enumerated above. The first concerns knowledge tracing: future work with full access to EdNet's item-level answer key could combine accuracy-based knowledge tracing, using contemporary deep-learning approaches [17], with the engagement-inequality findings reported here, testing directly whether the modest fluency effect observed in Section 4.6 corresponds to genuine mastery gains among retained users rather than to mere repeated exposure. The second concerns generalizability: because our module-level structural analysis was constrained by the small number of module presentations available in OULAD ($n = 22$), a pooled analysis across multiple institutions' learning analytics datasets would substantially increase statistical power to test whether the assessment-structure associations reported here replicate beyond a single institution and a single historical period. The third, and most directly actionable, concerns prospective validation: we recommend a study that operationalizes the three-question design framework proposed in Section 5.5, retention, structure, and equity, as a checklist applied during the design of a new AI-supported course, including courses that incorporate generative AI along the lines discussed in Section 5.6, with outcomes compared against a matched course designed without the checklist, so as to test whether the framework itself improves engagement and narrows the equity gaps documented here, extending the fairness-auditing methodology already applied to OULAD and to United States institutions [19, 20, 21] into a prospective, design-stage setting rather than confining it to retrospective audit.

8. Conclusion

Bringing together a conventional virtual-learning-environment dataset and an AI-tutoring interaction dataset provides converging evidence that behavioral engagement, rather than the presence of automation or adaptivity on its own, is the dominant signal associated with student success, and that this signal is distributed with marked inequality along both socio-economic and behavioral dimensions. For instructional designers and educational technologists tasked with integrating AI, generative or adaptive, into teaching and learning, these findings argue for prioritizing design-time simulation of course-structure changes grounded in established instructional design theory, engagement- and equity-aware evaluation of any AI personalization initiative, and onboarding and early-retention design as prerequisites without which adaptive AI has little opportunity to help the students who need it most. Addressing these priorities in the sequence proposed here serves directly the quality-education aims of Sustainable Development Goal 4 that motivate this Collection, and offers instructional designers a route to AI integration grounded in evidence about where existing course structures already succeed and fail, rather than in the technical possibilities of AI considered in isolation from that evidence.

Declarations

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Competing interests

The authors declare no competing interests.

Ethics

This study is a secondary analysis of two pre-existing, publicly available, fully anonymized datasets (OULAD and EdNet-KT1); no new data were collected from human participants by the author, and no identifiable information is contained in either source dataset.

Data availability

OULAD is publicly available under a Creative Commons Attribution 4.0 licence via the UCI Machine Learning Repository (<https://doi.org/10.24432/C5KK69>) and the Open University (https://analyse.kmi.open.ac.uk/open_dataset) [4]. EdNet-KT1 is publicly available under a Creative Commons Attribution-NonCommercial 4.0 licence from Riiid (<https://github.com/riiid/ednet>) [5]. Derived summary statistics, figures, and analysis code supporting this article are available from the corresponding author on reasonable request.

Author contributions

Opiyo Michael Ouru conceived the study, conducted the data analysis, and wrote the manuscript. Kenneth Goga Riary reviewed and refined the manuscript.

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Figures

Final Outcome Distribution (OULAD, N = 28,785)

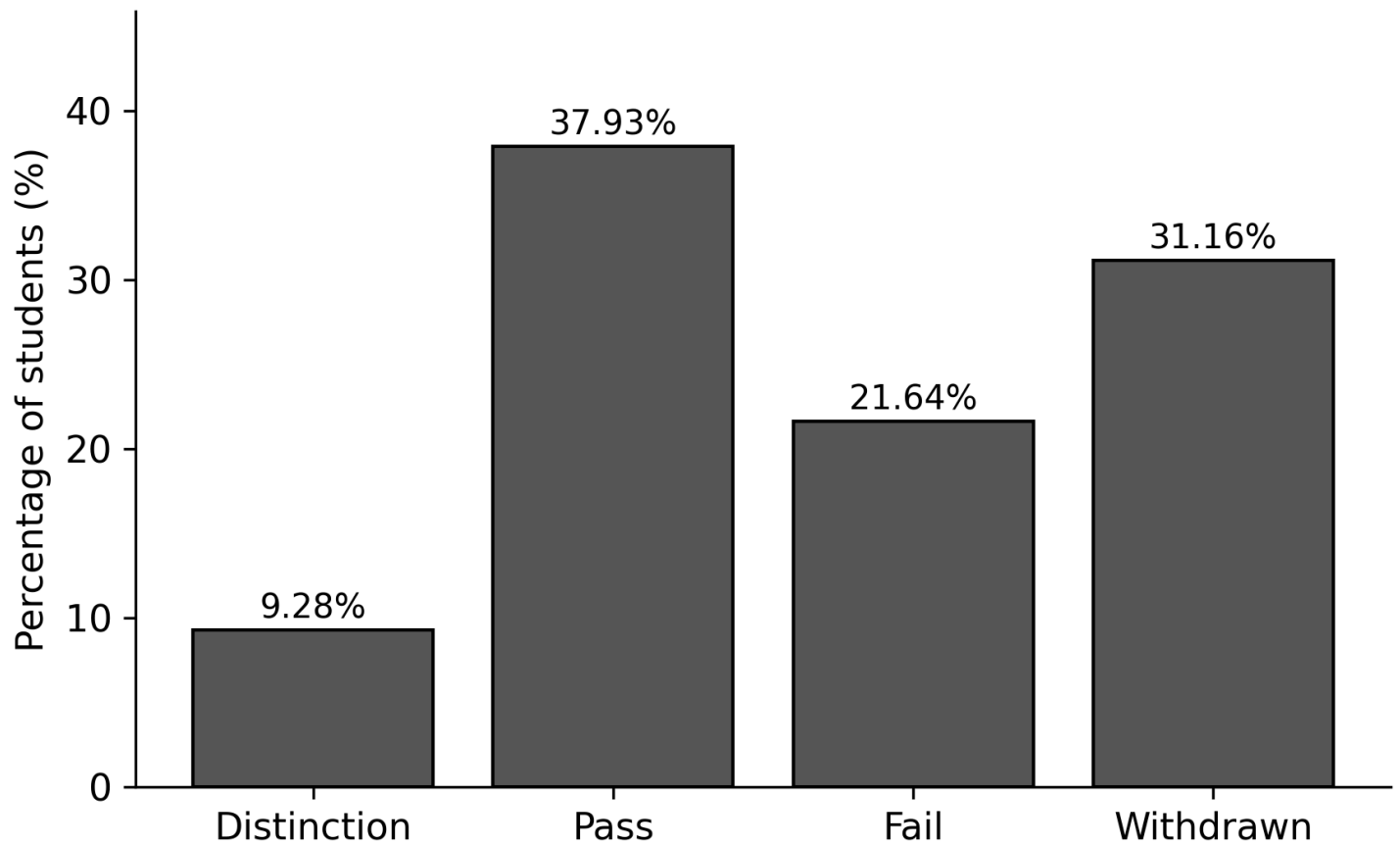


Figure 1

Final outcome distribution among 28,785 OULAD students, shown as a percentage of the full sample

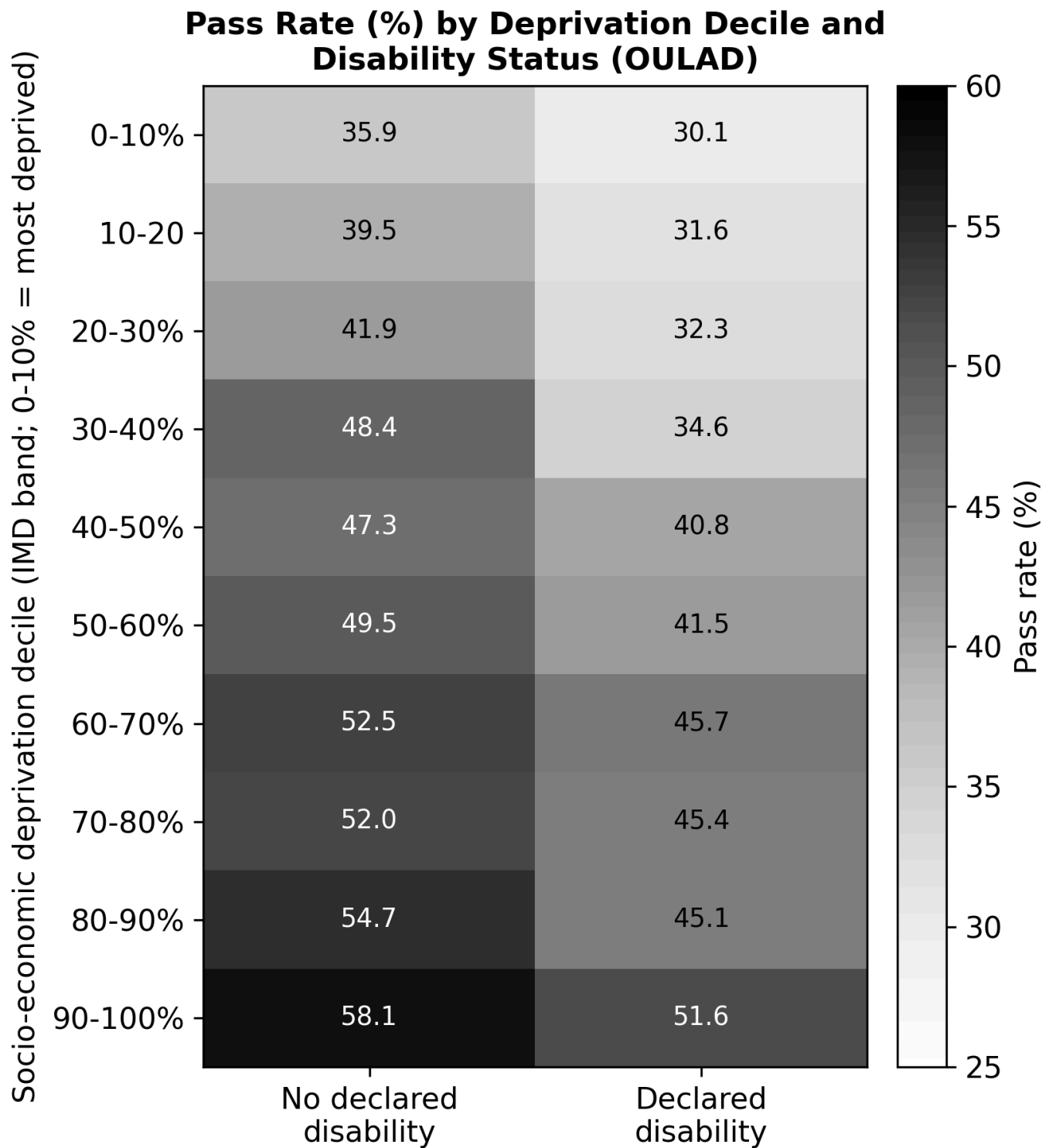


Figure 2

Pass rate (%) jointly by socio-economic deprivation decile and disability status, OULAD. Darker cells indicate higher pass rates; the leftmost column (no declared disability) is darker than the rightmost column (declared disability) at every deprivation decile, indicating that the two forms of disadvantage compound rather than operate independently

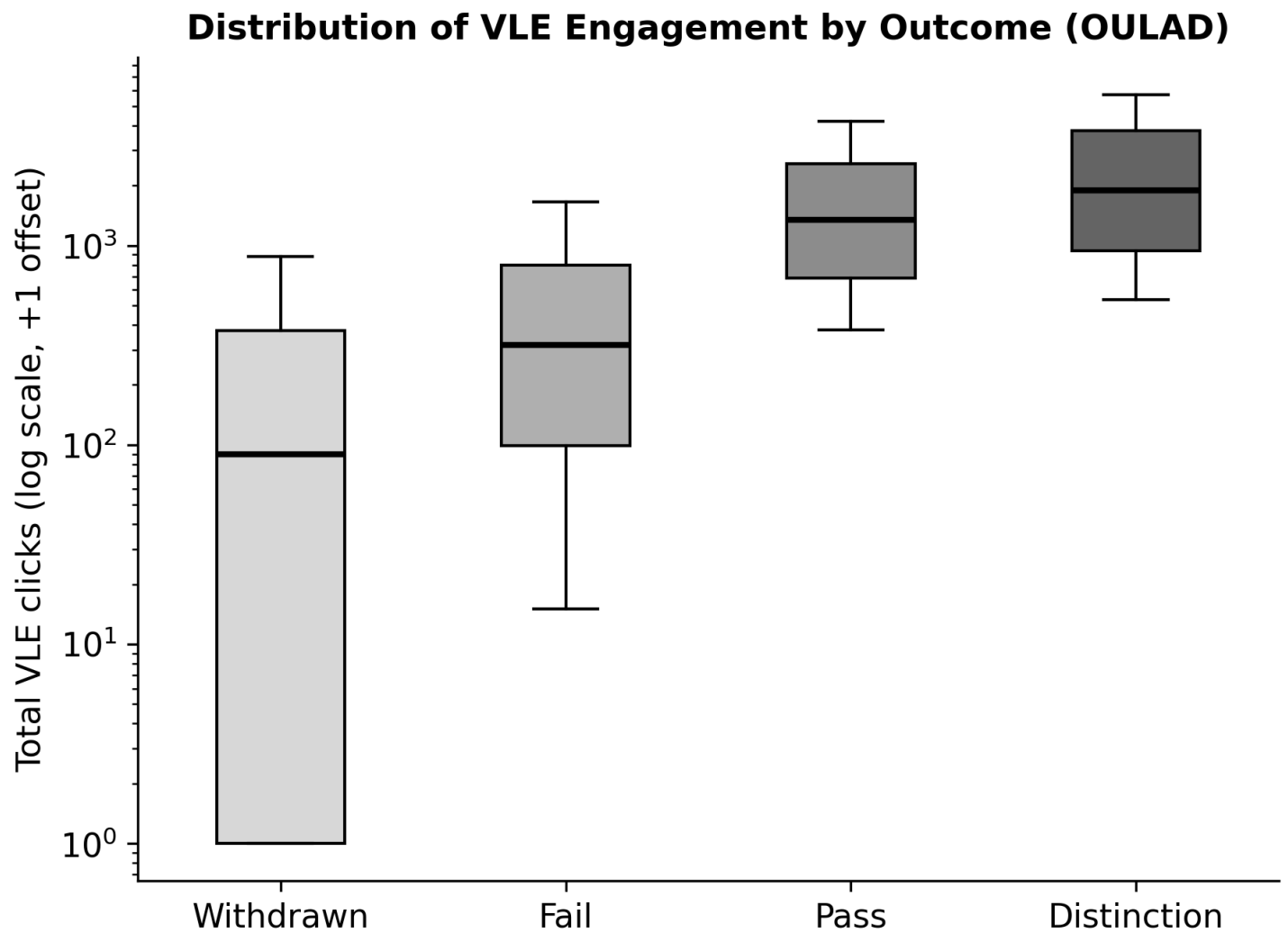


Figure 3

Distribution of total virtual learning environment clicks by final outcome, OULAD, shown on a logarithmic scale (values offset by one to accommodate zero-click students). Boxes span the interquartile range, the horizontal line marks the median, and whiskers extend to the 10th and 90th percentiles

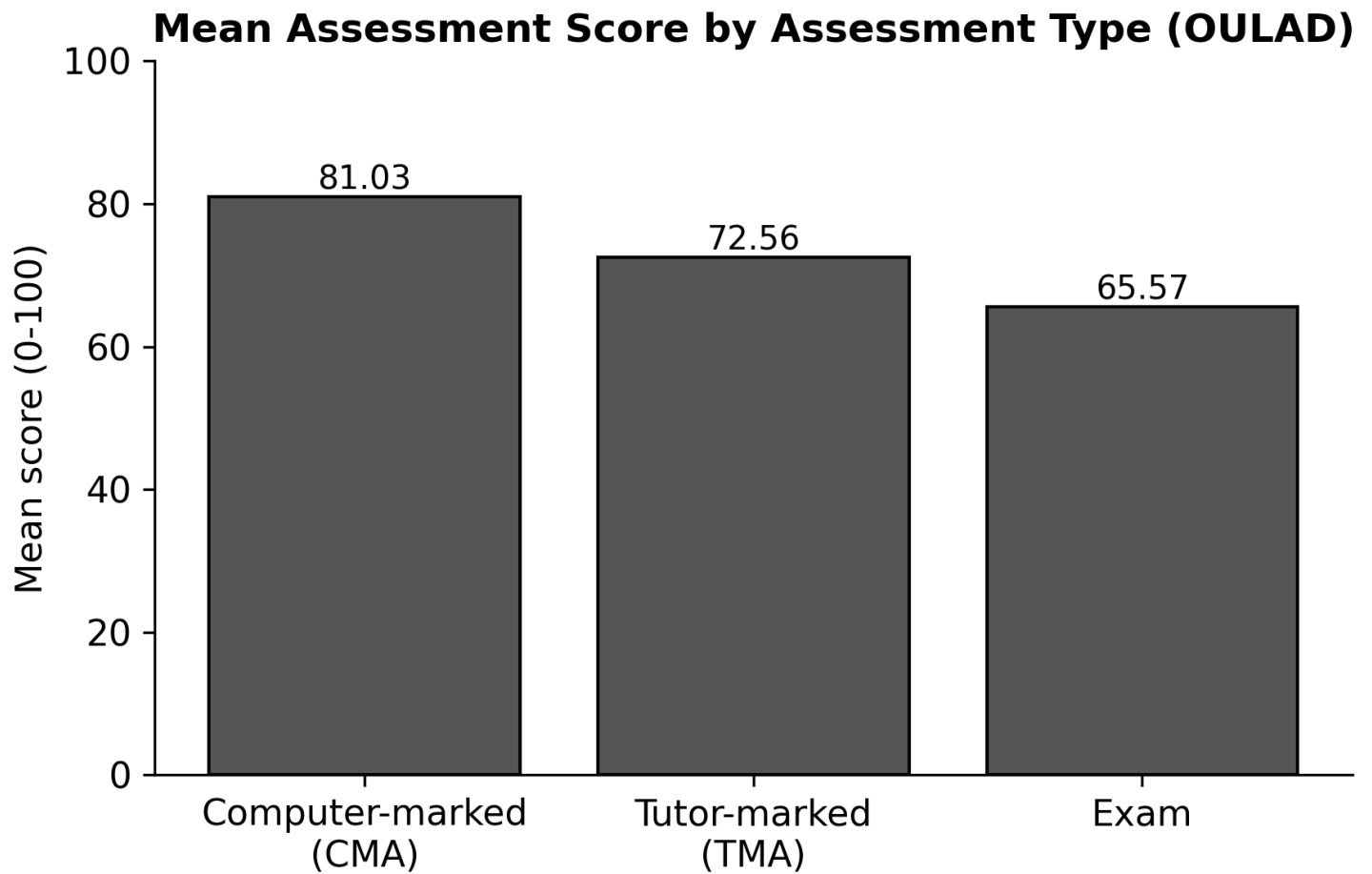


Figure 4

Mean assessment score by assessment type, OULAD

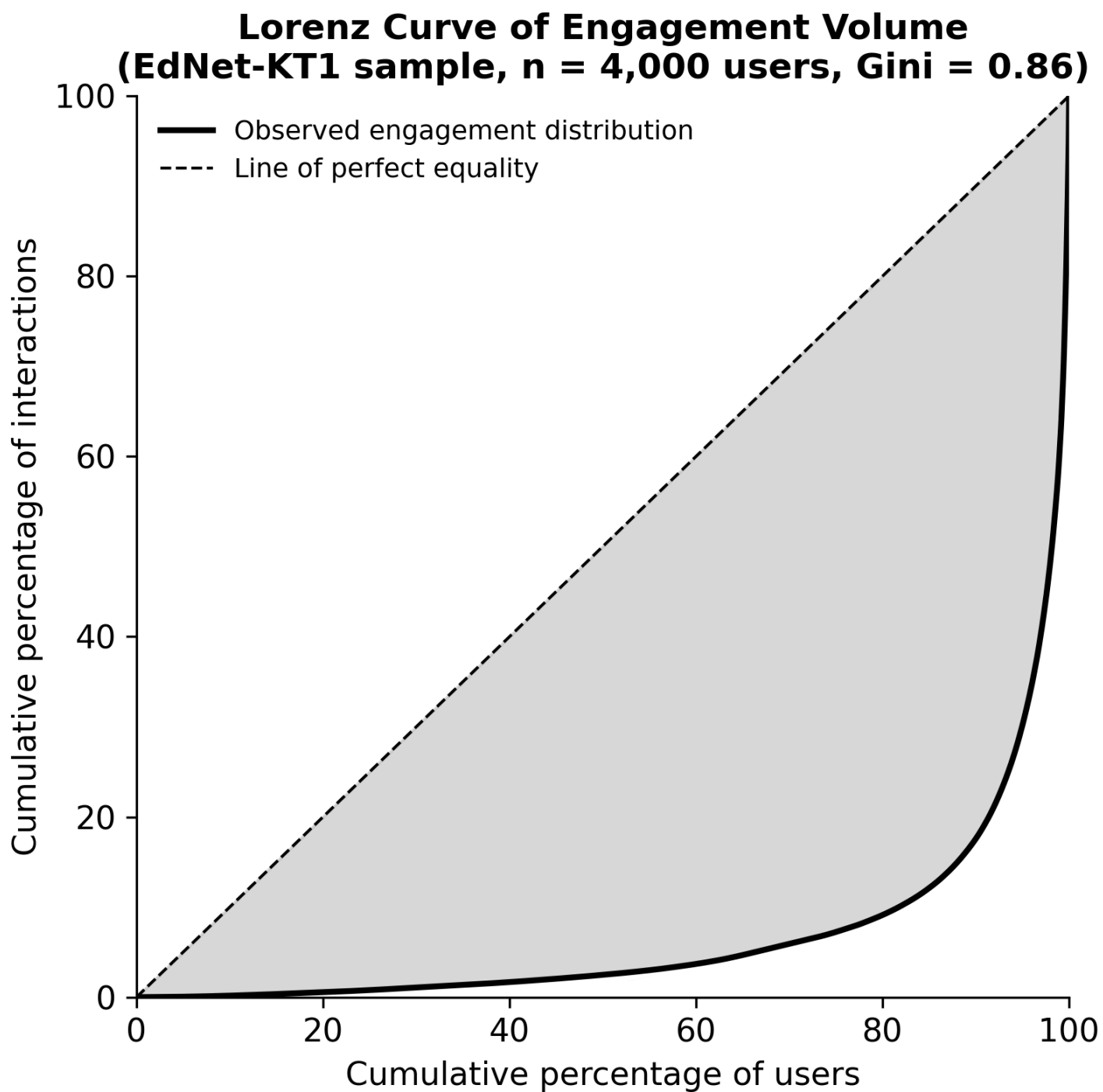


Figure 5

Lorenz curve of engagement volume among 4,000 sampled EdNet-KT1 users. The shaded area reflects the gap between observed engagement inequality (Gini = 0.86) and perfect equality

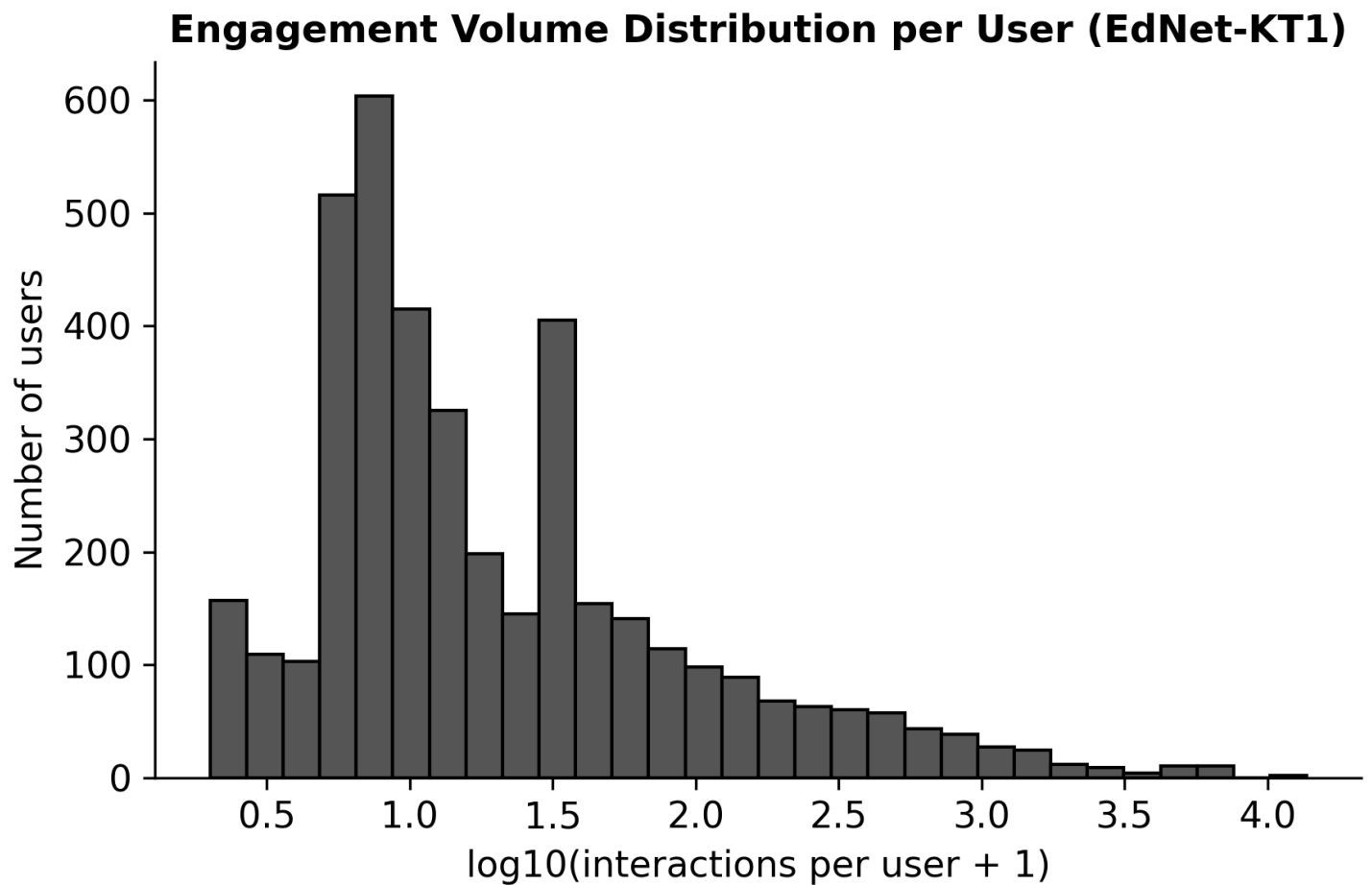


Figure 6

Distribution of engagement volume (interactions per user, log-transformed) among 4,000 sampled EdNet-KT1 users, illustrating extreme right skew